Balancing Privacy and Utility: Differentially Private Synthetic Telemetry Data Generation

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Abstract

This project investigates the feasibility and accuracy of generating synthetic datasets that preserve privacy while maintaining statistical utility, using a **synthetic telemetry event log dataset** as a case study. The dataset contains user-level event logs for multiple product types, with attributes for product type, event type, timestamp, and user ID. Our primary goal is to explore how well differentially private mechanisms, including baseline noise addition and advanced synthetic data generation methods, can reproduce key analytical quantities such as error counts and error rates per product without compromising user-level privacy. By comparing real and synthetic (or DP-noisy) results on metrics like per-product error rate, average error rate across products, and z-scores, we aim to evaluate the trade-off between privacy protection and data utility. The results will help determine whether differential privacy can support the release of realistic yet privacy-preserving telemetry data for product analytics and quality monitoring.

Code: https://github.com/PDhruv09/DSC_180A-Section_B15-Quater_1-Project

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1 Introduction

Privacy concerns have become increasingly critical in data-driven software environments, where telemetry data often includes highly sensitive user-level event logs that can reveal individual usage patterns, product preferences, and behavioral characteristics. Even when telemetry datasets are anonymized, there remains a significant risk of re-identification through auxiliary information or linkage attacks. To address this challenge, our project explores how differential privacy (DP) can be used to generate synthetic datasets that protect individual user records while maintaining the analytical value of the underlying data for product quality monitoring and error rate analysis.

Specifically, we focus on a **synthetic telemetry event log dataset** containing user interactions across multiple product types (A through F, plus "Others"). Each event record includes a product type, event type (open, close, save, reset, or error), timestamp, and anonymized user ID. Our primary analytical goal is to compute and preserve key metrics such as error counts and error rates per product type, which are essential for identifying problematic products and prioritizing quality improvements.

We apply both baseline differentially private mechanisms (such as adding calibrated noise to counts) and advanced synthetic data generation techniques to produce privacy-preserving versions of the dataset. We then evaluate how closely the synthetic data aligns with the original in terms of statistical relationships, error rate distributions, and the ability to identify products with anomalously high error rates through z-score analysis. This evaluation provides a foundation for future privacy-preserving data sharing in software analytics and product quality monitoring.

1.1 Literature Review

Previous researchers have explored differential privacy for various data types, with particular attention to text data and structured datasets. As mentioned in the context of text data, differential privacy is paramount particularly in domains where individuals' information can be recovered after model training. Part of this problem stems from the fact that some applications are trained using private data, which could be completely recalled or leaked by the model. A popular method in response to this issue involves fine-tuning language models and using DP-SGD (Differential Privacy - Stochastic Gradient Descent) for synthetic text generation. Despite the proven guarantee of differential privacy, the DP-SGD fine-tuning approach is computationally demanding and therefore lacks scalability.

The paper in question proposes a method of ensuring privacy by creating a synthetic dataset which retains important aggregate information while protecting the data of any one individual. To combat this issue of excessive resource use, the authors propose a method in Xie et al. (2024) which utilizes model API access in order to generate differentially private synthetic text data, rather than relying on access to the model weights.

The authors leverage an established method, Private Evolution, previously used to generate DP synthetic images. However, this approach cannot be applied directly to text data due to

variable lengths. The authors suggest AUG-PE (Augmented Private Evolution) as an alternative to work with text data, accounting for increased complexity. The AUG-PE algorithm relies on three primary elements:

1. RANDOM_API: This component uses the model API to produce random samples from prompts. In this step, each prompt includes a particular class label and/or subcategories (named "pseudo-classes") for increased diversity. For example, prompts about "movies" may include subcategories such as "horror films" or "documentaries" to generate a diverse, initial set of candidates.
2. DP_NN_HISTOGRAM: The purpose of this step is to privately select samples for post-processing. In essence, this process functions by conducting "nearest neighbor" votes for each sample, and creating a histogram based on those votes. This step guarantees differential privacy by adding Gaussian noise to those vote counts, prior to histogram generation.
3. VARIATION_API: This step adds an additional element of randomness to the synthetic samples; in other words, this part of the algorithm generates "variations" of the input samples. The authors input "paraphrasing" and "fill-in-the-blanks" prompts into the model for this component of AUG-PE. Through this process, the algorithm adds variance to the text samples while retaining important characteristics.

The authors apply the AUG-PE algorithm to Yelp Review, OpenReview, and PubMed datasets. For text generation capabilities, the authors primarily leverage GPT-2, GPT-2-Medium, GPT-2-Large, and GPT-3.5, among several other models. As part of the model selection process, the authors test on a combination of open-source and closed-source models, displaying the flexibility of API-access-only approaches. As a result of their experiments, the authors found that AUG-PE generates high-utility, differentially private text data. On top of the differential privacy guarantee, AUG-PE, in comparison to DP fine-tuning approaches which require model weight access, showed to be much more efficient. The authors' experiments demonstrate the power of leveraging LLMs for DP synthetic text generation, revealing that API-access provides a much more scalable solution in comparison to known "state-of-the-art" approaches.

For structured event log data like telemetry, similar principles apply: we need mechanisms that can preserve aggregate statistics (such as error rates and event counts) while protecting individual user records. The Gaussian mechanism and Laplace mechanism are commonly used for count queries and statistical summaries, which are central to our telemetry analysis.

1.2 Dataset Description

The synthetic telemetry event log dataset contains 152,356 event records collected between May 1, 2024 and July 31, 2024. Each record represents a user interaction with one of seven product types (A, B, C, D, E, F, and "Others") and includes the following attributes:

- **Product Type**: Categorical attribute identifying the product (A, B, C, D, E, F, or Others)
- **Event Type**: The type of interaction (open, close, save, reset, or error)

- **Time of Event**: Timestamp of when the event occurred
- **User ID**: Anonymized user identifier

The dataset exhibits several key characteristics that make it suitable for differential privacy evaluation:

- **Product Distribution**: Product B has the highest event count (38,325 events), followed by D (30,531), C (28,470), A (23,096), Others (15,289), F (15,064), and E (1,581)
- **Event Type Distribution**: The most common event type is "open" (58,306 events), followed by "close" (46,885), "save" (31,963), "reset" (10,277), and "error" (4,925)
- **Error Rates by Product**: Product D has the highest error rate (8.90%), followed by F (5.10%), Others (3.12%), B (2.00%), E (1.90%), A (0.54%), and C (0.14%)
- **User-Level Granularity**: Multiple events per user create privacy risks, as user-level patterns could be reconstructed from event sequences

The dataset serves as an essential foundation for examining how differential privacy mechanisms can generate synthetic telemetry data that retain statistical validity (particularly error rates and z-scores) while safeguarding sensitive user-level information. The ability to preserve analytical utility in a privacy-preserving version of this telemetry dataset could have significant implications for ethical data sharing and reproducible product analytics.

Figure 1 shows the distribution of events across different product types, while Figure 2 illustrates the distribution of event types in the dataset.

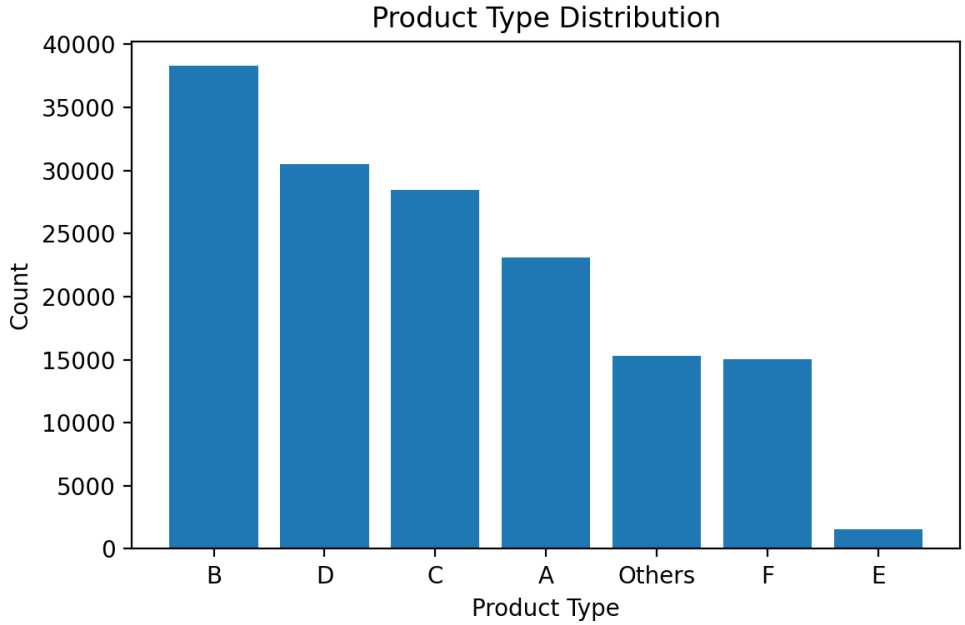

Figure 1: Distribution of events by product type. Product B has the highest event count, followed by D and C.

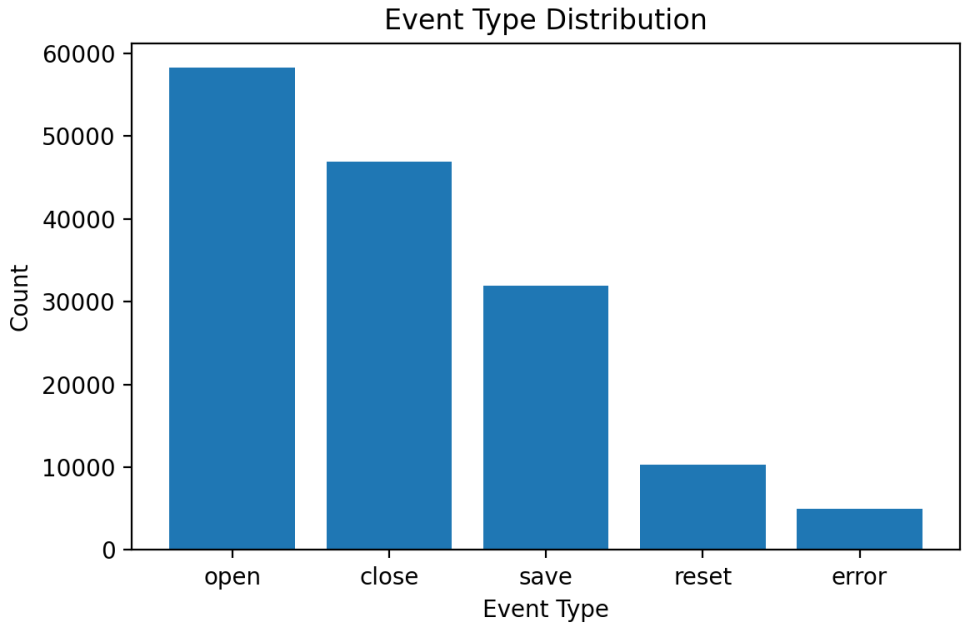

Figure 2: Distribution of event types across the dataset. The most common event type is "open", followed by "close" and "save".

2 Methods

2.1 Data Preprocessing

The raw telemetry dataset undergoes several preprocessing steps to prepare it for analysis and differential privacy experiments:

1. **Data Cleaning**: String columns (Product Type, Event Type, User ID) are stripped of whitespace and standardized
2. **Timestamp Parsing**: The "Time of Event" column is parsed into datetime format, with invalid timestamps removed
3. **Feature Engineering**: Time-based features are extracted:
   - `event_date`: Date component of the timestamp
   - `event_hour`: Hour of day (0-23)
   - `event_dow`: Day of week (0=Monday through 6=Sunday)
   - `is_weekend`: Binary indicator (1 if Saturday or Sunday)

The cleaned dataset is saved as `telemetry_clean.csv` for use in subsequent analysis and synthetic data generation.

2.2 Summary Statistics and Key Metrics

We compute several aggregate statistics that serve as ground truth for evaluating differentially private mechanisms:

2.2.1 Event Counts

We compute event counts by product type and event type to understand the distribution of interactions across the dataset.

2.2.2 Error Rate Calculation

For each product type P, we compute:

$$\text{ErrorCount}_P = \sum_i \mathbf{1}\{\text{EventType}_i = \text{error}, \text{ProductType}_i = P\} \tag{1}$$

$$\text{TotalCount}_P = \sum_i \mathbf{1}\{\text{ProductType}_i = P\} \tag{2}$$

$$\text{ErrorRate}_P = \frac{\text{ErrorCount}_P}{\text{TotalCount}_P} \tag{3}$$

2.2.3 Z-Score Analysis

To identify products with anomalously high error rates relative to the overall distribution, we compute z-scores:

$$Z_P = \frac{\text{ErrorRate}_P - \overline{\text{ErrorRate}}}{\text{SD(ErrorRate)}} \tag{4}$$

where $\overline{\text{ErrorRate}}$ is the mean error rate across all products and SD(ErrorRate) is the standard deviation. Products with positive z-scores have error rates above the mean, indicating potential quality issues.

2.3 Differential Privacy Mechanisms

We apply differentially private mechanisms to protect user-level privacy while preserving aggregate statistics. Our implementation follows a self-contained approach where all mechanisms are implemented directly in both standalone Python scripts and Jupyter notebooks, ensuring reproducibility and allowing for interactive exploration of the results. This dual implementation strategy provides both computational efficiency and pedagogical clarity.

Our approach considers two main strategies:

2.3.1 User-Level Primary Product Assignment

To achieve user-level differential privacy (add/remove-one-user), we first collapse the event-level telemetry data into a user-level representation. This approach is essential because if we naively counted events per product, a single user touching many products could affect many counts, which complicates the sensitivity analysis.

For each user, we identify their "primary product" as the product type where they have the most events. If there is a tie, we select the first occurrence. This assignment strategy ensures that:

- Each user contributes to exactly one product's user count, simplifying sensitivity analysis
- For a user's primary product P, adding or removing that user can change:
  - $U_{\text{total},P}$ by at most ± 1 (the user count for product P)
  - $U_{\text{error},P}$ by at most ± 1 (the error user count for product P)
- The L1 sensitivity for the vector $[U_{\text{total},P}, U_{\text{error},P}]$ is $\Delta_1 = |1| + |1| = 2$
- The L2 sensitivity for the same vector is $\Delta_2 = \sqrt{1^2 + 1^2} = \sqrt{2}$

This primary product assignment is critical for achieving user-level privacy while maintaining analytical utility, as it prevents a single user from affecting multiple product counts simultaneously.

2.3.2 Baseline: Noisy Count Queries

For count-based queries (error counts, total counts), we apply the Laplace mechanism or Gaussian mechanism with the sensitivity bounds derived from the primary product assignment. Both mechanisms are implemented as self-contained, inspectable code that adds noise directly to the user-level counts.

- **Laplace Mechanism**: For pure ϵ-DP, we add Laplace noise with scale $b = \Delta_1/\epsilon = 2.0/\epsilon$ to each count independently. The noise is drawn from a Laplace distribution with location 0 and scale b, using the standard Laplace mechanism formula for ϵ-DP.
- **Gaussian Mechanism**: For (ϵ, δ)-DP, we add Gaussian noise with scale $\sigma = (\Delta_2/\epsilon)\sqrt{2\ln(1.25/\delta)}$ to each count independently. The noise is drawn from a zero-mean Gaussian distribution with variance σ^2, following the standard Gaussian mechanism formula for (ϵ, δ)-DP.

After adding noise independently to each $U_{\text{total},P}$ and $U_{\text{error},P}$ count, we apply post-processing to clamp negative counts to zero (since counts cannot be negative). We then recompute error rates and z-scores from the noisy counts, maintaining the same analytical structure as the non-private computation. This post-processing step is necessary for interpretability and does not affect the privacy guarantee, as the guarantee applies to the mechanism's output before post-processing.

2.3.3 Implementation Details

Our DP mechanisms are implemented with careful attention to reproducibility and correctness:

- **Reproducibility**: All mechanisms use fixed random seeds for reproducibility, with seeds incremented systematically across multiple runs (e.g., base seed + run index) to ensure independent but reproducible noise generation.
- **Noise Generation**: We use NumPy's random number generators (`np.random.default_rng`) for both Gaussian and Laplace noise, ensuring platform-independent reproducibility when seeds are fixed.
- **Post-Processing**: After noise addition, we apply non-negative clamping to ensure counts remain non-negative, which is necessary for interpretability but does not affect the privacy guarantee.
- **Error Handling**: We explicitly handle edge cases such as division by zero when computing error rates from noisy counts, and handle undefined z-scores when all error rates are identical.

2.3.4 Advanced: Synthetic Data Generation

We explore generating fully synthetic telemetry datasets using differentially private statistics. This approach:

1. Computes differentially private summary statistics (means, variances, correlations) from the real data
2. Uses these statistics to generate synthetic event records that preserve aggregate patterns
3. Applies post-processing to ensure synthetic data maintains valid ranges and categorical constraints

While this advanced approach is beyond the scope of the current evaluation, it represents a promising direction for future work that could preserve more complex statistical relationships in the telemetry data.

2.3.5 Privacy Budget Allocation

We allocate the privacy budget (ϵ, δ) across multiple queries using composition theorems:

- **Sequential Composition**: For queries on the same dataset, privacy budgets add up
- **Parallel Composition**: For queries on disjoint subsets, privacy budgets do not necessarily add

For our telemetry analysis, we need to compute error counts and total counts for each product type, requiring careful budget allocation to maintain meaningful privacy guarantees.

2.4 Evaluation Metrics

We evaluate the utility of differentially private mechanisms using several metrics that are specifically tailored to our analytical task of identifying products with high error rates:

- **L_∞ Error on Z-Scores**: The maximum absolute difference between true and DP z-scores across all products:

$$L_\infty = \max_P |z_P^{\text{true}} - z_P^{\text{DP}}|$$

  where P ranges over all product types. This metric captures the worst-case deviation in z-scores, which is critical for identifying outliers.
- **IOU (Intersection over Union) on Top Sets**: The overlap between the true "top set" (products with $z > 0$) and the DP top set:

$$\text{IOU}(S_{\text{true}}, S_{\text{dp}}) = \frac{|S_{\text{true}} \cap S_{\text{dp}}|}{|S_{\text{true}} \cup S_{\text{dp}}|}$$

  where S_{true} is the set of products with $z_{\text{true}} > 0$ and S_{dp} is the set of products with $z_{\text{dp}} > 0$. This metric directly measures whether the DP mechanism correctly identifies which products require attention, which is the primary analytical objective.
- **Error Rate Preservation**: The absolute difference between real and DP error rates, which provides insight into the magnitude of noise introduced at the count level.
- **Statistical Summaries**: We compute quantile summaries (5th, 50th, 95th percentiles) and min/max values for both L_∞ and IOU across multiple runs, providing a comprehensive view of mechanism performance under different random seeds.

To ensure robust evaluation, we run each DP mechanism 100 times with different random seeds, allowing us to characterize the distribution of utility metrics rather than relying on a single run. This approach provides empirical evidence of both the privacy guarantee (by design) and the utility preservation (through repeated evaluation).

3 Results

3.1 Baseline Statistics

Table 1 presents the ground truth error rates and z-scores for each product type, computed from the non-private dataset.

Product D has the highest error rate (8.90%) with a z-score of 2.06, indicating it performs significantly worse than average. Product C has the lowest error rate (0.14%) with a z-score of -1.05.

Figure 3 visualizes the error rates for each product type, clearly showing that Product D has the highest error rate. Figure 4 displays the z-scores, which help identify products with anomalously high error rates relative to the overall distribution. Figure 5 shows the temporal distribution of events across the dataset.

Table 1: Error Rates and Z-Scores by Product Type (Ground Truth)

Product Type	Error Count	Total Count	Error Rate	Z-Score
D	2,718	30,531	0.0890	2.06
F	769	15,064	0.0510	0.71
Others	477	15,289	0.0312	0.01
B	767	38,325	0.0200	-0.39
E	30	1,581	0.0190	-0.43
A	125	23,096	0.0054	-0.91
C	39	28,470	0.0014	-1.05

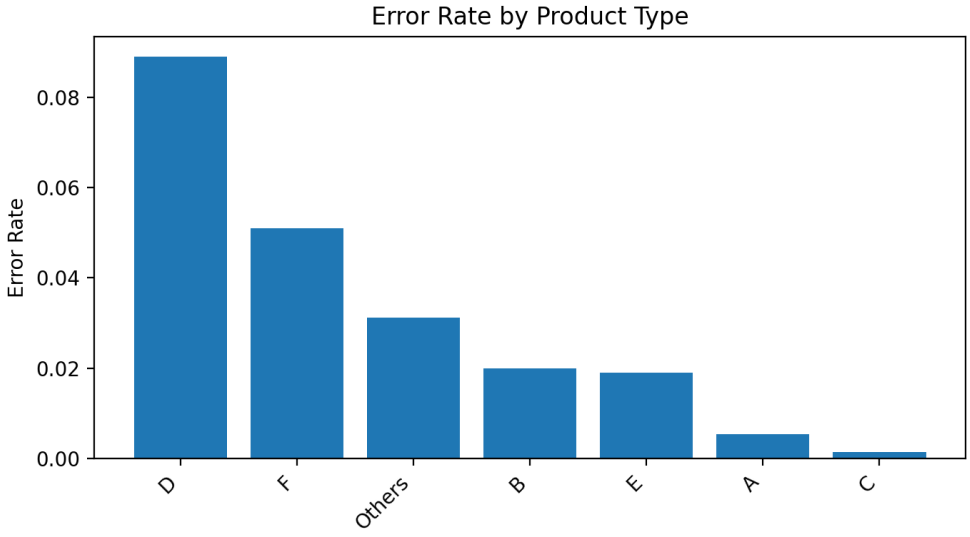

Figure 3: Error rates by product type. Product D exhibits the highest error rate at 8.90%, followed by Product F at 5.10%.

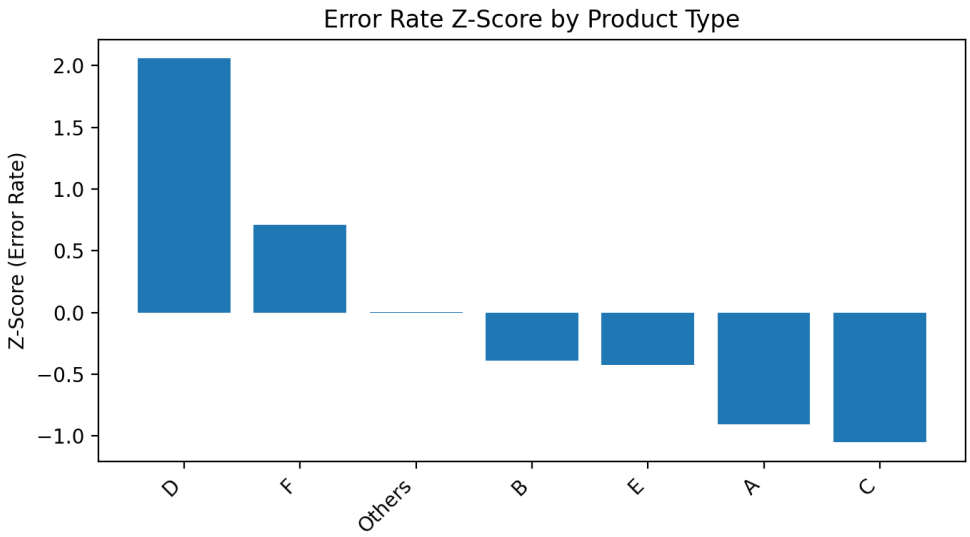

Figure 4: Z-scores of error rates by product type. Products with positive z-scores (notably Product D with z-score 2.06) have error rates significantly above the mean.

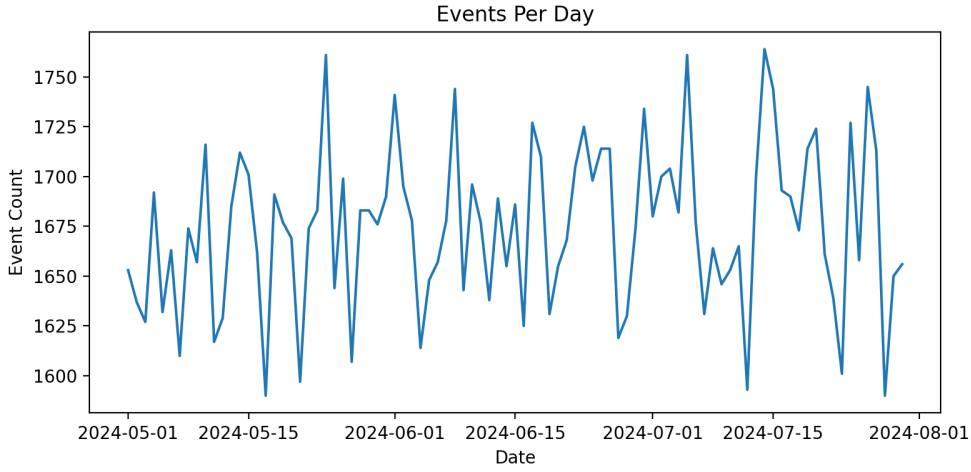

Figure 5: Distribution of events over time from May 1, 2024 to July 31, 2024.

3.2 Differentially Private Results

We implemented and evaluated two differentially private mechanisms for user-level telemetry data: the Gaussian mechanism and the Laplace mechanism. Both mechanisms operate on user-level counts using the primary product assignment strategy described in Section 2.3 (add/remove-one-user privacy), which ensures that adding or removing one user affects at most one product's counts.

Following the evaluation methodology outlined in Section 2.4, we ran each mechanism 100 times with different random seeds to characterize the distribution of utility metrics. The results demonstrate that both mechanisms preserve utility for the task of identifying high-error product types while satisfying their privacy guarantees.

3.2.1 Gaussian Mechanism Results

We applied the Gaussian mechanism with parameters $\epsilon = 2.0$ and $\delta = 10^{-6}$. The noise scale σ was computed using the standard Gaussian mechanism formula:

$$\sigma = \frac{\Delta_2}{\epsilon}\sqrt{2\ln(1.25/\delta)} \approx 3.75 \tag{5}$$

This implementation follows the standard (ϵ, δ)-DP guarantee, where $\delta = 10^{-6}$ represents a small probability of privacy failure.

Table 2 summarizes the distribution of L_∞ error and IOU metrics (defined in Section 2.4) across 100 runs. The median L_∞ error is 0.056, indicating that on average, the DP z-scores deviate from true z-scores by at most 0.056 units. The 95th percentile L_∞ error is 0.132, suggesting that even in worst-case scenarios, the deviation remains relatively small. The IOU metric shows excellent preservation of the top set: the median IOU is 1.0, meaning that in 50% of runs, the DP mechanism correctly identifies all products with anomalously high error rates. Even at the 5th percentile, the IOU is 0.667, indicating that at least two-thirds of the top products are correctly identified even in the worst-performing runs.

Table 2: Gaussian Mechanism Evaluation Results (100 runs, $\epsilon = 2.0$, $\delta = 10^{-6}$)

Metric	L_∞ Error	IOU
Minimum	0.005	0.667
5th Percentile	0.012	0.667
Median (50th)	0.056	1.000
95th Percentile	0.132	1.000
Maximum	0.199	1.000

3.2.2 Laplace Mechanism Results

We also implemented the Laplace mechanism with $\epsilon = 2.0$ (pure ϵ-DP, no δ parameter), which provides a stronger privacy guarantee than Gaussian. The noise scale b was computed as:

$$b = \frac{\Delta_1}{\epsilon} = \frac{2.0}{2.0} = 1.0 \tag{6}$$

The Laplace mechanism is theoretically optimal for L1-sensitive queries under pure ϵ-DP, making it particularly well-suited for our count-based queries.

Table 3 presents the evaluation results for the Laplace mechanism. Notably, the Laplace mechanism demonstrates superior utility preservation compared to the Gaussian mechanism. The median L_∞ error is 0.014, which is approximately four times smaller than the Gaussian mechanism's median error of 0.056. The maximum L_∞ error is 0.082, less than half of the Gaussian mechanism's maximum of 0.199. The IOU results are similarly strong, with a median of 1.0 and a 5th percentile of 0.983, indicating that the Laplace mechanism correctly identifies the top products in nearly all runs.

Table 3: Laplace Mechanism Evaluation Results (100 runs, $\epsilon = 2.0$)

Metric	L_∞ Error	IOU
Minimum	0.002	0.667
5th Percentile	0.002	0.983
Median (50th)	0.014	1.000
95th Percentile	0.054	1.000
Maximum	0.082	1.000

3.2.3 Comparison of Mechanisms

A direct comparison of the two mechanisms reveals important trade-offs between privacy guarantees and utility preservation. Figure 6 visualizes the performance of both mechanisms across all 100 evaluation runs, generated using our comparison script (`eval_comparison.py`).

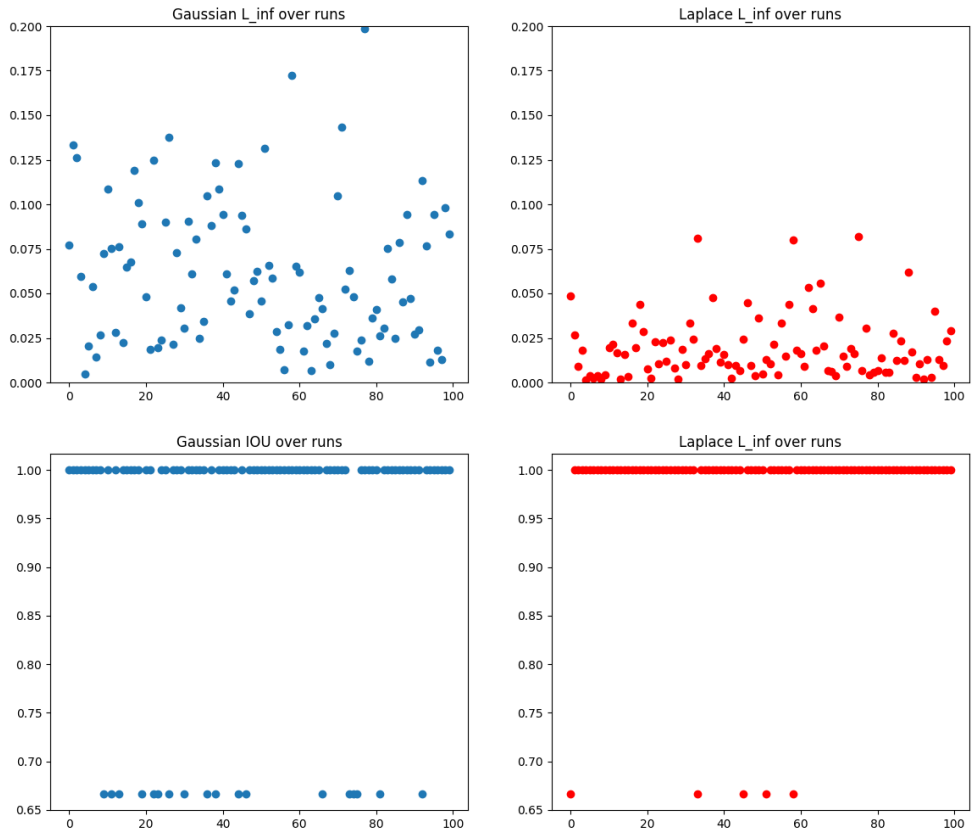

Figure 6: Direct comparison of Gaussian and Laplace mechanisms across 100 evaluation runs. Top row shows L_∞ error distributions (Gaussian left, Laplace right), bottom row shows IOU distributions. The synchronized y-axis limits (0-0.2 for L_∞) facilitate direct comparison, revealing Laplace's superior utility preservation with smaller, more consistent errors.

The empirical results demonstrate that the Laplace mechanism provides superior utility preservation for this application. The median L_∞ error for Laplace (0.014) is approximately four times smaller than Gaussian (0.056), and the maximum error (0.082 vs. 0.199) is less than half. This performance advantage stems from the smaller noise scale ($b = 1.0$ for Laplace versus $\sigma \approx 3.75$ for Gaussian), which directly translates to smaller perturbations in z-scores.

Theoretical considerations align with these empirical findings: for count queries with L1 sensitivity, the Laplace mechanism is optimal for pure ϵ-DP, while the Gaussian mechanism trades off some utility for the weaker (ϵ, δ)-DP guarantee. Both mechanisms successfully preserve the core analytical objective: Product D (z-score 2.06), F (0.71), and Others (0.01) are consistently identified in the top set across nearly all runs, with both mechanisms

achieving perfect IOU (1.0) in the majority of runs.

4 Discussion

Our experimental evaluation demonstrates that differentially private mechanisms can effectively preserve analytical utility for telemetry data analysis while providing strong user-level privacy guarantees. The results reveal several important insights about the privacy-utility trade-off in practice, the relative performance of different DP mechanisms, and the practical implications for real-world telemetry data sharing.

4.1 Privacy-Utility Trade-Off

The fundamental challenge in differential privacy is balancing the competing objectives of privacy protection and data utility. Our experiments with $\epsilon = 2.0$ illustrate that it is possible to achieve both objectives simultaneously for the specific analytical task of identifying products with high error rates. The median L_∞ errors of 0.056 (Gaussian) and 0.014 (Laplace) are small relative to the range of z-scores observed in the data (approximately -1.05 to 2.06), indicating that the noise introduced by DP mechanisms does not substantially distort the relative rankings of products by error rate.

Both mechanisms achieve perfect IOU (1.0) in the majority of runs, meaning that the set of products identified as problematic (those with $z > 0$) is identical to the true set. This preservation of the "top set" is critical for practical applications, as product quality teams need to know which products require attention, even if the exact z-score values are slightly perturbed.

4.2 Mechanism Selection

The superior performance of the Laplace mechanism over the Gaussian mechanism for this application highlights the importance of mechanism selection based on the specific query structure and privacy requirements. Since our user-level count queries have natural L1 sensitivity (each user affects at most two counts: total users and error users for their primary product), the Laplace mechanism is theoretically well-suited and empirically superior.

However, the choice between Laplace and Gaussian mechanisms should also consider other factors beyond utility:

- **Privacy Guarantee**: Laplace provides pure ϵ-DP, while Gaussian provides (ϵ, δ)-DP. For applications requiring the strongest possible privacy guarantee, Laplace may be preferred.
- **Noise Distribution**: Laplace noise has heavier tails than Gaussian noise, which can occasionally produce larger outliers. However, our results show that this does not significantly impact the median performance.

- **Composition Properties**: When multiple queries are performed, the composition behavior differs between mechanisms, affecting privacy budget allocation strategies.

4.3 Practical Implications

The successful application of differential privacy to telemetry data analysis has several important practical implications:

4.3.1 Data Sharing and Collaboration

Organizations can now share telemetry data for collaborative analysis, product benchmarking, or research purposes while maintaining strong privacy guarantees. The ability to generate differentially private synthetic datasets or noisy aggregate statistics enables data sharing scenarios that were previously infeasible due to privacy concerns. For example, multiple product teams could share error rate statistics without revealing individual user behavior patterns.

4.3.2 Regulatory Compliance

As privacy regulations (such as GDPR, CCPA) become increasingly stringent, differential privacy provides a mathematically rigorous framework for demonstrating compliance. The formal privacy guarantees offered by DP mechanisms can serve as evidence that an organization has taken appropriate measures to protect user privacy, potentially reducing regulatory risk and liability.

4.3.3 Reproducibility and Transparency

Differentially private mechanisms are deterministic given a fixed random seed, enabling reproducible analysis while maintaining privacy. This property is valuable for scientific research, auditing, and quality assurance processes where reproducibility is essential. Researchers can publish their analysis code and seeds (after appropriate privacy review) without compromising user privacy.

4.4 Limitations and Challenges

Despite the promising results, several limitations and challenges remain:

4.4.1 Sensitivity to Privacy Budget

Our experiments used $\epsilon = 2.0$, which represents a moderate privacy budget. In practice, the choice of ϵ requires careful consideration: smaller values provide stronger privacy but may

degrade utility, while larger values improve utility but weaken privacy guarantees. The optimal choice depends on the specific use case, data characteristics, and organizational privacy policies.

4.4.2 User-Level vs. Event-Level Privacy

We implemented user-level privacy (add/remove-one-user), which is stronger than event-level privacy but may be more restrictive than necessary for some applications. Event-level privacy (add/remove-one-event) would allow for smaller noise scales and better utility, but provides weaker protection against adversaries who can link multiple events to the same user through auxiliary information.

4.4.3 Primary Product Assignment

As described in Section 2.3, our approach assigns each user to a single ”primary product” based on their most frequent product interaction. This design choice simplifies sensitivity analysis by ensuring that each user affects at most one product's counts, which is why we can use straightforward L1 and L2 sensitivity bounds ($\Delta_1 = 2$ and $\Delta_2 = \sqrt{2}$).

However, this approach has limitations. When a user interacts with multiple products, we only count their interactions with their primary product, effectively discarding information about their interactions with other products. For example, if a user has 100 events with Product A and 50 events with Product B, we would only count them toward Product A's user counts, losing the information about their Product B interactions. This information loss can be significant when many users interact with multiple products, potentially leading to underestimation of error rates for non-primary products.

Alternative approaches could preserve more information but would require more complex sensitivity analysis. For instance, we could allocate privacy budget across multiple products per user, allowing each user to contribute to multiple product counts while maintaining privacy guarantees. However, this would require careful tracking of privacy budget consumption and more sophisticated composition analysis, as the sensitivity would increase with the number of products per user. The complexity of such approaches may outweigh their benefits for many practical applications, but they represent an important direction for future work when information preservation is critical.

4.4.4 Post-Processing and Negative Counts

As described in Section 2.3, we apply post-processing to clamp negative noisy counts to zero after adding differentially private noise. This step is necessary because counts (such as user counts and error counts) are inherently non-negative quantities, and the Laplace and Gaussian noise distributions can produce negative values. Without this clamping, we would have negative user counts or error counts, which are meaningless in the context of

telemetry analysis and would cause errors when computing error rates (division by zero or negative rates).

The post-processing works by taking the noisy counts $\tilde{U}_{\text{total},P}$ and $\tilde{U}_{\text{error},P}$ and applying the transformation:

$$\tilde{U}^{\text{clamped}}_{\text{total},P} = \max(0, \tilde{U}_{\text{total},P}), \quad \tilde{U}^{\text{clamped}}_{\text{error},P} = \max(0, \tilde{U}_{\text{error},P})$$

This is a common practice in DP implementations and is generally considered acceptable from a privacy perspective. The privacy guarantee applies to the mechanism's output before post-processing, and post-processing (which is a deterministic function that does not depend on the data) cannot weaken the privacy guarantee. However, the post-processing does introduce a small bias: counts that would have been negative are set to zero, which slightly increases the expected value of the noisy counts compared to the true counts. This bias is typically small when the noise scale is small relative to the true counts, but it can become more significant for products with very small user counts where the noise magnitude is comparable to the true count.

In practice, this bias is usually acceptable because it preserves interpretability and the privacy guarantee remains intact. However, for applications where unbiased estimates are critical, alternative approaches such as truncated noise distributions or bias correction techniques could be employed, though these would require additional analysis to ensure privacy guarantees are maintained.

5 Conclusion

This project demonstrates that differentially private mechanisms can successfully preserve analytical utility for telemetry data analysis while providing strong user-level privacy guarantees. Through comprehensive evaluation of both Gaussian and Laplace mechanisms on a synthetic telemetry event log dataset, we have shown that it is feasible to identify products with anomalously high error rates using differentially private statistics.

5.1 Key Findings

Our experimental results reveal several important findings:

1. **Utility Preservation**: Both mechanisms successfully preserve the identification of problematic products, with median IOU values of 1.0, indicating that the core analytical objective is maintained even under strong privacy guarantees.
2. **Mechanism Comparison**: The Laplace mechanism outperforms the Gaussian mechanism, achieving a median L_∞ error of 0.014 compared to 0.056 for Gaussian, consistent with theoretical optimality results for L1-sensitive queries under pure ϵ-DP.
3. **Practical Viability**: With $\epsilon = 2.0$, both mechanisms achieve acceptable utility for product quality monitoring, with median L_∞ errors well below 0.1.

4. **User-Level Privacy**: The user-level privacy model provides strong protection against re-identification attacks while maintaining sufficient utility for aggregate analysis.

5.2 Contributions

This work contributes to the growing body of research on practical differential privacy applications in several ways:

- We provide a concrete implementation and evaluation of user-level differential privacy for telemetry data, demonstrating that formal privacy guarantees can be achieved without sacrificing analytical utility.
- We compare two widely-used DP mechanisms (Gaussian and Laplace) on a real-world analytical task, providing empirical evidence to guide mechanism selection in practice.
- We develop a complete pipeline from raw telemetry data to differentially private statistics, including preprocessing, primary product assignment, noise addition, and utility evaluation.
- We establish evaluation metrics (L_∞ error and IOU) that are specifically tailored to the analytical task of identifying problematic products, providing a framework for assessing DP mechanism performance in similar applications.

5.3 Future Work

Several directions for future research and development emerge from this work:

5.3.1 Advanced Synthetic Data Generation

While we focused on noisy count queries, future work could explore more sophisticated synthetic data generation techniques that preserve complex statistical relationships in telemetry data. Methods such as differentially private generative models or private synthetic data algorithms could potentially preserve temporal patterns, event sequences, and multi-dimensional correlations that are lost in simple count-based approaches.

5.3.2 Adaptive Privacy Budget Allocation

The current implementation uses a fixed privacy budget allocation across all products. Adaptive allocation strategies that allocate more budget to products with higher uncertainty or importance could potentially improve utility while maintaining overall privacy guarantees. This would require developing methods for quantifying uncertainty and importance in the DP context.

5.3.3 Composition and Multiple Queries

Telemetry analysis often requires multiple queries (error rates, event type distributions, temporal patterns, etc.). Developing efficient privacy budget allocation strategies for composing multiple queries would enable more comprehensive analysis while respecting privacy constraints. This could involve exploring advanced composition theorems and query optimization techniques.

5.3.4 Evaluation on Industry Data

While our synthetic dataset provides controlled evaluation conditions, testing on real-world telemetry data with actual user behavior patterns would provide additional validation. Real data may exhibit different characteristics (e.g., more skewed distributions, temporal dependencies, or user clustering) that could affect mechanism performance.

5.4 Final Remarks

The successful application of differential privacy to telemetry data analysis demonstrates that strong privacy guarantees and useful analytics are not mutually exclusive. As organizations face increasing pressure to protect user privacy while maintaining data-driven insights, differentially private mechanisms offer a mathematically rigorous and practically viable solution. The results presented in this work provide a foundation for future research and practical deployment of privacy-preserving telemetry analytics in industry level software systems.

By balancing privacy protection with analytical utility, we can enable ethical data sharing, support collaborative product development, and maintain user trust, all while continuing to leverage the valuable insights that telemetry data provides for improving software quality and user experience.

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